

Stage 4 — Causal checks and the bootstrap correction

Where the Shortcut Sits: supervisor meeting 4 of 6

Report generated from the repository on September 28, 2026

In one sentence. The real-artifact crossover survives covariate matching and regression adjustment in all three cohorts where matching is feasible (A2: 3 of 3 supported); moving the same real artifact inside and outside the ROI of the same image produces a crossover too, but a neutral paste of the same shape reproduces most of it, so only a small part — significant for hair and thyroid calipers — is attributable to the artifact’s content. The bootstrap correction widened intervals by a median factor of 1.56 and changed 179 of 2466 significance verdicts.

Notation

$Y \in \{0, 1\}$	diagnosis (1 = melanoma / malignant / solid-component tumour / erosion).
$A \in \{0, 1\}$	artifact present (hair, sonographer caliper, capsule debris, chest tube).
ROI	region of interest (lesion, nodule, tumour, lesion box, lungs); drawn in green ; artifacts in red .
r	overlap $ \text{artifact} \cap \text{ROI} / \text{artifact} $. Trap A: $r \geq 0.5$ (in-ROI); Trap B: $r < 0.1$ (out-of-ROI).
Environments	training and correlated test $P(A=1 Y=1) = 0.9$, $P(A=1 Y=0) = 0.1$; reversed test 0.1/0.9; clean test: artifact independent of Y (real traps: half of each class; sweeps: absent).
ERM, mask	linear head on the frozen features of the original image / of the ROI-masked image.
Crossover	$[\text{mask} - \text{ERM}]_{\text{out}} - [\text{mask} - \text{ERM}]_{\text{in}}$ of reversed-test AUROC (real traps: Trap B – Trap A; sweeps and transplant: location interaction on the same images).
CI	95% crossed seed×image bootstrap interval (Stage 1, Sec. ??); every interval in these reports uses it.
$ \Delta p $	same-head counterfactual sensitivity: change of the predicted probability when the artifact is inserted.
SMD	standardised mean difference between two groups.

1 Where we are

Stage 3 showed a positive crossover with real artifacts. Real artifacts are not placed at random: in dermoscopy, hair lies on large lesions more often than on small ones; in thyroid ultrasound, calipers inside the nodule are drawn on larger, more heavily measured nodules. Artifact-based debiasing studies in dermoscopy make the same point about where artifacts occur (Bissoto et al., 2020; Nauta et al., 2022). The crossover could therefore reflect *which images* carry in-ROI artifacts rather than *where* the artifact lies. This meeting reports the three checks designed to separate the two — matching, regression adjustment on the matched sample (A2, new in this round) and

transplantation of the same artifact with a paste-edge control — and, because it concerns what every interval means, the full consequences of the bootstrap correction.

2 Question

- Q4a** Is the crossover explained by measured differences between in-ROI and out-of-ROI artifact images?
- Q4b** Does it survive when the artifact is moved within the *same* image, and how much of that is due to the artifact rather than to pasting something into the image?
- Q4c** Which earlier claims change when the uncertainty is estimated correctly?

Why three checks Matching removes measured imbalance but depends on the chosen covariates; regression on the matched sample (a doubly robust combination) protects against residual imbalance and model misspecification of either step alone; transplantation removes image-level confounding altogether but introduces its own artifact — the paste. The neutral paste (same mask shape, same compositing, filled with the donor’s own clean tissue) measures that paste effect. Rejected alternatives: inverse-probability weighting (extreme weights with standardised differences above 2 in dermoscopy and capsule) and synthetic control images (they would test Stage 2 again, not real artifacts).

3 Methods

Balance and matching (R1) Pre-specified covariates per cohort (dermoscopy: lesion area, hair amount, age, sex, site, source, image size; thyroid: nodule area, caliper pixels, image size, grey level, nodule position; ovary and capsule analogous). Standardised mean differences (SMD) between Trap A and Trap B artifact images; greedy 1:1 nearest-neighbour matching on the logit of a propensity model within label, without replacement, calliper 0.2 SD (0.05 SD as a stricter sensitivity analysis, R7). A cohort with fewer than 30 matched images per label is infeasible.

Regression adjustment (A2) For each reversed-test image of the matched sample, the *placement value* of the masked minus the ERM score (for a positive: the weighted share of negatives ranked below it; for a negative: the complement of the share of positives ranked above it), so that the class means reproduce the AUROC and the unadjusted difference between traps equals the crossover exactly (verified per seed). Within each seed, weighted least squares of this outcome on an intercept, the Trap-B indicator, the label and all matching covariates; the adjusted crossover is the Trap-B coefficient averaged over seeds. Crossed bootstrap with placement values and regression recomputed in every replicate (2,000 replicates); Holm over the three cohorts.

Transplant (R2) and neutral paste (R5) For each artifact-free recipient image, a real artifact instance is cut from a donor of the same cohort (the donor’s mask pixels, largest connected component) and alpha-composited at a position fully inside and at a position fully outside the recipient’s ROI, with a random dihedral transform. The trap environments are then built as in Stage 1 with the pasted instance as the artifact. The neutral paste keeps the mask shape, the transform and the compositing but fills the mask with tissue from the donor image at a window that does not touch its artifact.

Bootstrap correction (A1) Every archived interval with a regenerated counterpart was compared row by row: old and new interval, whether “excludes zero” changed, width ratio and point-estimate difference.

4 Experiments

Table 1: Experiments of this stage. Commit times come from this machine and are not independent proof of order; the final registration was additionally pushed to GitHub before the runs.

Experiment	Status	Registration	First commit (local)	Support criterion	
Covariate balance (SMD) and matched traps, M1	supported registered	REVIEW2 (R1)	29cd0da, 18:56	2026-09-27	matched crossover CI > 0, Holm over 3 feasible cohorts
Stricter calliper 0.05 SD (R7)	supported registered	REVIEW3	85da735, 23:43	2026-09-27	sensitivity, descriptive
Real-artifact transplant, T1	supported registered	REVIEW2 (R2)	29cd0da, 18:56	2026-09-27	interaction CI > 0, Holm over 4
Neutral paste (paste-edge control), N3	supported registered	REVIEW3 (R5)	85da735, 23:43	2026-09-27	artifact — neutral interaction CI > 0
Regression-adjusted crossover, A2	supported registered	FINAL (A2)	c45d482, 00:35	2026-09-28	adjusted crossover CI > 0, Holm over 3
Crossed bootstrap everywhere, A1	supported registered	FINAL (A1)	c45d482, 00:35	2026-09-28	rewrite claims whose CI now includes 0
Capsule matched analysis	not supported descriptive	REVIEW2 (R1)	29cd0da, 18:56	2026-09-27	infeasible under the registered rule

5 Results

5.1 Imbalance and matching

The in-ROI and out-of-ROI artifact images differ strongly (Table 2): in dermoscopy the largest SMD is 2.127 (lesion area: hair lies inside large lesions), in capsule 2.574 (roi area). Matching brings the maximum SMD to 0.108 (dermoscopy), 0.045 (thyroid) and 0.079 (ovary); capsule matching is infeasible (19 pairs in the smaller label, SMD after matching 0.231). The crossover is essentially unchanged by matching (Table 3): dermoscopy +0.154 [+0.123, +0.186], thyroid +0.238 [+0.186, +0.290], ovary +0.163 [+0.050, +0.273]; with the stricter calliper +0.124 [+0.086, +0.162], +0.218 [+0.165, +0.270] and +0.152 [+0.041, +0.265]. Capsule, matched descriptively, gives +0.230 [+0.036, +0.434] — lower than the unmatched +0.377 [+0.327, +0.426] and imprecise.

Table 2: Balance of pre-specified covariates between the in-ROI (Trap A) and out-of-ROI (Trap B) artifact images, before and after 1:1 nearest-neighbour matching within label (calliper 0.2 SD, pooled standardised mean differences). Registered rule: fewer than 30 matched images per label in either trap makes a cohort infeasible; target max |SMD| ≤ 0.1 after matching.

Cohort	Least balanced covariate	max SMD before	after matching	min matched pairs per label	feasible (rule)
ISIC hair	lesion_area	2.127	0.108	398	yes
Thyroid	caliper_px_log	0.527	0.045	93	yes
Ovary	caliper_px_log	0.389	0.079	86	yes
Capsule	roi_area	2.574	0.231	19	no

Table 3: Crossover before and after covariate matching (reversed-test AUROC, crossed 95% CIs). The 0.05 SD calliper is the stricter sensitivity analysis of the third review (R7).

Cohort	Crossover, all images	matched (0.2 SD)	matched (0.05 SD)	M1 (Holm, 3)
ISIC hair	+0.147 [+0.111, +0.185]	+0.154 [+0.123, +0.186]	+0.124 [+0.086, +0.162]	supported
Thyroid	+0.239 [+0.196, +0.280]	+0.238 [+0.186, +0.290]	+0.218 [+0.165, +0.270]	supported
Ovary	+0.170 [+0.087, +0.252]	+0.163 [+0.050, +0.273]	+0.152 [+0.041, +0.265]	supported
Capsule	+0.377 [+0.327, +0.426]	+0.230 [+0.036, +0.434]	+0.328 [+0.087, +0.573]	descriptive (infeasible)

Table 4: Per-covariate balance, thyroid and ovary (pooled over labels). The dermoscopy and capsule tables (17 and 20 covariates) are in `results/rerun_2026-09-28/review2/R1_balance.csv`.

Cohort	Covariate	SMD before	SMD after
Thyroid	roi_area	+0.395	<i>n/a</i>
Thyroid	caliper_px_log	+0.527	<i>n/a</i>
Thyroid	width	−0.143	<i>n/a</i>
Thyroid	height	−0.123	<i>n/a</i>
Thyroid	mean_grey	+0.183	<i>n/a</i>
Thyroid	roi_centroid_row	+0.060	<i>n/a</i>
Ovary	roi_area	+0.320	<i>n/a</i>
Ovary	caliper_px_log	+0.389	<i>n/a</i>
Ovary	width	−0.298	<i>n/a</i>
Ovary	height	+0.047	<i>n/a</i>
Ovary	mean_grey	+0.107	<i>n/a</i>
Ovary	subtype=0.0	+0.149	−0.079
Ovary	subtype=1.0	+0.059	+0.072
Ovary	subtype=2.0	−0.159	−0.012
Ovary	subtype=3.0	−0.054	+0.000
Ovary	subtype=4.0	−0.022	+0.024
Ovary	subtype=6.0	+0.024	+0.056
Ovary	subtype=7.0	−0.020	−0.062

5.2 Regression adjustment on the matched sample (A2)

The adjusted crossover equals the unadjusted one to the third decimal in every cohort (Table 5): dermoscopy +0.156 [+0.127, +0.187] (unadjusted +0.154 [+0.124, +0.186]), thyroid +0.238 [+0.184, +0.291] (+0.238 [+0.185, +0.291]), ovary +0.165 [+0.056, +0.273] (+0.163 [+0.058, +0.268]); Holm-adjusted $p = 0.002$, 0.002 and 0.002 . All 3 pre-registered tests are supported. Residual imbalance after matching (up to the per-label maxima in Table 5) does not carry the effect. Capsule stays descriptive, as registered. What this cannot exclude is an *unmeasured* difference between the images.

Table 5: Regression-adjusted crossover on the matched traps (A2): per-image placement values of the masked minus the ERM score regressed on the Trap-B indicator, the label and all matching covariates, within seed; crossed bootstrap (2,000 replicates). Capsule is not analysed (matching infeasible; descriptive only).

Cohort	Test images per seed	Covariate columns	Post-matching max SMD pooled / per label	Unadjusted crossover	Adjusted (doubly robust)	p (Holm, 3)	A2
ISIC hair	2960	17	0.108 / 0.164	+0.154 [+0.124, +0.186]	+0.156 [+0.127, +0.187]	0.002	supported
Thyroid	2150	6	0.045 / 0.150	+0.238 [+0.185, +0.291]	+0.238 [+0.184, +0.291]	0.002	supported
Ovary	576	11	0.079 / 0.140	+0.163 [+0.058, +0.268]	+0.165 [+0.056, +0.273]	0.002	supported

5.3 The same artifact inside and outside the ROI

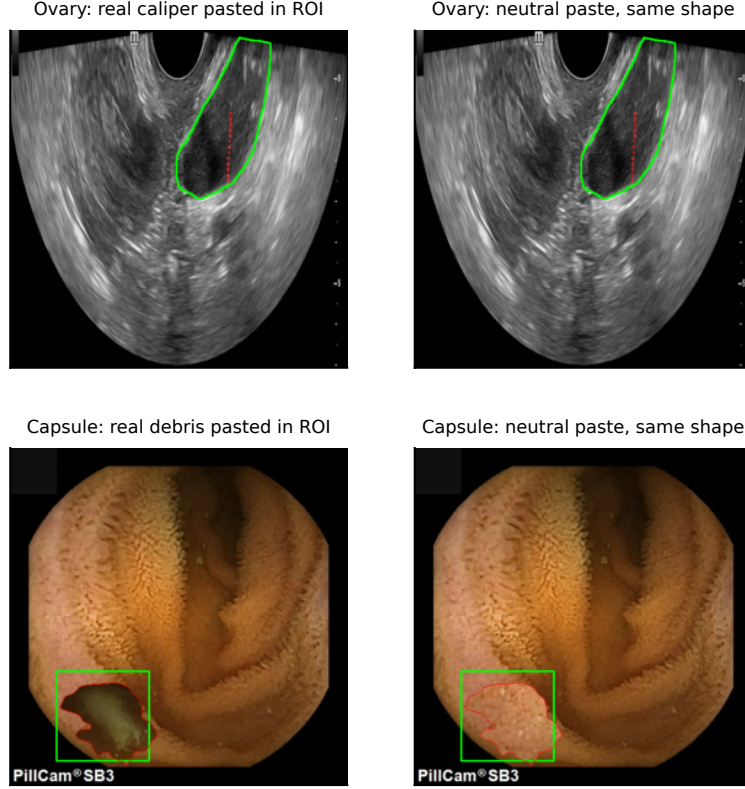


Figure 1: Transplant and neutral paste on the same recipient images (openly licensed cohorts; MMOTU, SEE-AI CC BY 4.0). Left: a real artifact instance pasted inside the ROI (green; the pasted region outlined in red). Right: the neutral paste — same mask shape and compositing, filled with artifact-free tissue from the donor.

Pasting the same real instance outside versus inside the ROI yields a positive location interaction in every cohort (Table 6; dermoscopy $+0.253$ [$+0.214, +0.293$], thyroid $+0.277$ [$+0.256, +0.298$], ovary $+0.074$ [$+0.044, +0.104$], capsule $+0.497$ [$+0.427, +0.567$]). The neutral paste, however, reproduces 52% to 92% of that interaction (Table 7): pasting *anything* into the ROI changes what masking does, because the paste itself is a new, label-correlated pattern inside the ROI (the environments make the pasted region label-correlated whatever it contains). Only the difference between the two — N3, the interaction with the real artifact minus the interaction with the neutral paste — is attributable to the artifact’s content: dermoscopy $+0.049$ [$+0.026, +0.072$], thyroid $+0.132$ [$+0.116, +0.148$], ovary $+0.022$ [$-0.002, +0.048$] and capsule $+0.037$ [$-0.017, +0.093$] (2 of 4 excluding zero). **Consequently no harm figure from the transplant (for example, the effect of masking with the pasted hair inside the lesion) is attributed to the real artifact.** The transplant supports the *location* mechanism — a label-correlated pattern inside the ROI survives masking — and N3 shows that the real hair and caliper content adds to it.

Table 6: Real-artifact transplant (T1): the same real artifact instance, cut from a donor image, pasted outside and inside the ROI of the same artifact-free recipient images. These effects combine the artifact with the paste itself (seam, occluded tissue); Table 7 separates them.

Cohort	mask — ERM, pasted outside ROI	pasted inside ROI	Location interaction	Balanced: interaction
ISIC hair	$+0.178$ [$+0.128, +0.227$]	-0.075 [$-0.114, -0.036$]	$+0.253$ [$+0.214, +0.293$]	-0.052 [$-0.071, -0.033$]
Thyroid	$+0.349$ [$+0.321, +0.379$]	$+0.072$ [$+0.043, +0.100$]	$+0.277$ [$+0.256, +0.298$]	-0.016 [$-0.025, -0.007$]
Ovary	$+0.064$ [$+0.001, +0.126$]	-0.010 [$-0.075, +0.056$]	$+0.074$ [$+0.044, +0.104$]	-0.011 [$-0.025, +0.002$]
Capsule	$+0.322$ [$+0.262, +0.384$]	-0.175 [$-0.241, -0.109$]	$+0.497$ [$+0.427, +0.567$]	-0.032 [$-0.068, +0.004$]

Table 7: Paste-edge control (R5): the same mask shape and compositing, filled with the donor’s own artifact-free tissue. Only the N3 column — the interaction with the real artifact minus the interaction with the neutral paste — is attributable to the artifact’s content.

Cohort	mask – ERM, neutral outside	neutral inside	Neutral interaction	Artifact – neutral (N3)	Neutral / artifact interaction
ISIC hair	+0.129 [+0.079, +0.181]	−0.075 [−0.116, −0.035]	+0.204 [+0.164, +0.244]	+0.049 [+0.026, +0.072]	81%
Thyroid	+0.200 [+0.175, +0.226]	+0.055 [+0.028, +0.081]	+0.145 [+0.130, +0.161]	+0.132 [+0.116, +0.148]	52%
Ovary	+0.072 [+0.009, +0.135]	+0.019 [−0.041, +0.080]	+0.052 [+0.021, +0.084]	+0.022 [−0.002, +0.048]	71%
Capsule	+0.215 [+0.154, +0.280]	−0.244 [−0.314, −0.174]	+0.459 [+0.383, +0.535]	+0.037 [−0.017, +0.093]	92%

5.4 Consequences of the bootstrap correction

Across 2466 regenerated intervals with an archived counterpart, the corrected intervals are wider by a median factor of 1.56 (interquartile range 1.34–1.74; Figure 2); 147 lost significance and 32 gained it (Table 8). 160 regenerated point estimates differ from the archived ones by more than 0.03; most are in the capsule synthetic sweeps (up to 0.222), whose cohort was rebuilt on this machine: only 32 of the 71 archived artifact-bearing test frames fall in the new test set of 150 frames, because a seeded shuffle of the group list re-dealt the split. Table 9 lists the 33 verdict changes that touch a quoted claim. The claims rewritten as a result:

- Controlled sweeps: masking at 25% overlap (dermoscopy, DINOv2 at 518 and 224 px) and at 50% overlap (ovary) is no longer significantly harmful; MedSigLIP’s benefit at full thyroid overlap is no longer significant; the lesion-size strata lost two cells (Stage 2).
- Occlusion control: the small positive effects of masking on the 50/50 test are no longer significant; the conclusion (occlusion does not harm) is unchanged.
- Primary family: 11 instead of 12 of 16 supported after Holm (Stage 3); P3 ovary lost support.
- Natural test sets: masking’s gain on easy thyroid pairs and on MSK dermoscopy, and U-MtE’s gain over masking on hard thyroid pairs, are no longer significant (Stage 5).
- Fine-tuned models: several thyroid comparisons changed in both directions; fine-tuned results remain supplementary (Stage 5).
- Gained: masking’s cost in clean-test AUROC in the hair Trap A is now significant.

Table 8: Old versus corrected intervals by result group (every regenerated interval with an archived counterpart; `results/bootstrap_correction/sweeps_umte_old_vs_new.md`).

Result group	Intervals	Lost significance	Gained	$ \Delta $ estimate > 0.03	max $ \Delta $ estimate	median width ratio
PRIMARY_CLAIMS.csv	16.0	2.0	0.0	1.0	0.067	1.36
adhoc_bootstraps.json	45.0	1.0	0.0	6.0	0.076	1.54
analysis	4.0	0.0	0.0	0.0	0.015	1.63
capsule	390.0	22.0	9.0	21.0	0.057	1.58
finetune	42.0	4.0	3.0	9.0	0.062	1.21
natural	75.0	4.0	0.0	0.0	0.000	1.74
ovary	252.0	11.0	0.0	0.0	0.013	1.66
review2	60.0	1.0	0.0	0.0	0.000	1.63
review3	55.0	0.0	0.0	0.0	0.000	1.00
spec_e13	301.0	19.0	9.0	5.0	0.056	1.55
synthetic	786.0	73.0	7.0	102.0	0.222	1.47
thyroid	428.0	9.0	4.0	16.0	0.098	1.62
zero_shot	12.0	1.0	0.0	0.0	0.012	2.00

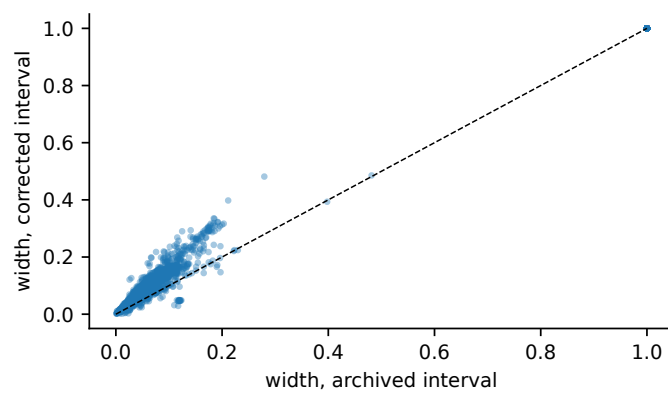


Figure 2: Width of each corrected interval against the archived one (dashed: equal width).

Table 9: Verdict changes among intervals that concern masking, the primary family, natural test sets, fine-tuning and the transplant. Only the corrected interval is shown; the archived interval is in the comparison file.

Result	Row	Corrected estimate [95% CI]	Significance
PRIMARY_CLAIMS.csv	P2 U-MtE (protected) > mask Capsule	−0.006 [−0.018, +0.006]	lost
PRIMARY_CLAIMS.csv	P4 U-MtE+balanced > balanced	+0.031 [−0.014, +0.077]	lost
adhoc_bootstraps.json	Ovary sweep capsule_dino518 r=1.0 umte_ balanced-balanced	+0.057 [−0.008, +0.121]	lost
thyroid/convnext384_universal	trapA mask clean mask erm	+0.021 [−0.000, +0.041]	lost
synthetic/istic2018/dino518_ruler_ fixed_corr_main	0.25 mask erm test_rev	−0.059 [−0.127, +0.008]	lost
synthetic/istic2018/dino518_ruler_ fixed_occlusion_occlusion	0.0 mask erm test_uncorr	+0.036 [−0.016, +0.086]	lost
synthetic/istic2018/dino518_ruler_ fixed_occlusion_occlusion	0.5 mask erm test_uncorr	+0.041 [−0.010, +0.090]	lost
synthetic/istic2018/dino518_ruler_ fixed_occlusion_occlusion	0.75 mask erm test_uncorr	+0.037 [−0.013, +0.086]	lost
synthetic/istic2018/dino518_ruler_ fixed_occlusion_occlusion	1.0 mask erm test_uncorr	+0.039 [−0.010, +0.091]	lost
synthetic/istic2018/dino224_ruler_ fixed_corr_main	0.25 mask erm test_rev	−0.049 [−0.117, +0.018]	lost
synthetic/thyroid/medsiglip448_ caliper_corr_main	1.0 mask erm test_rev	+0.044 [−0.011, +0.104]	lost
synthetic/ovary/dino518_caliper_ corr_main	0.5 mask erm test_rev	−0.107 [−0.269, +0.054]	lost
capsule/dino518_scale	trapA mask clean mask erm	+0.012 [−0.006, +0.031]	lost
spec_e13/dino518_matched05	trapA mask clean all mask erm	−0.026 [−0.053, +0.000]	lost
spec_e13/dino518_spec_scale	trapB mask clean all mask erm	+0.028 [−0.000, +0.055]	lost
spec_e13/dino518_spec_scale	trapA mask test_rev HAM mask erm	−0.044 [−0.090, +0.002]	lost
spec_e13/dino518_spec_scale	trapB mask clean all mask erm	−0.025 [−0.055, +0.003]	lost
spec_e13/dino518_spec_universal	trapA mask clean all mask erm	−0.023 [−0.045, −0.001]	gained
spec_e13/dino518_spec	trapA mask clean all mask erm	−0.023 [−0.045, −0.001]	gained
spec_e13/dino518_spec_ablation	trapA mask clean all mask erm	−0.023 [−0.045, −0.001]	gained
spec_e13/dino518_spec_splice_ v2	trapA mask clean all mask erm	−0.023 [−0.045, −0.001]	gained
natural/thyroid_dino518_repro/ natural_boot.json	hard mte-mask	+0.030 [+0.000, +0.061]	lost
natural/thyroid_dino518_repro/ natural_boot.json	easy mask-erm	+0.042 [−0.011, +0.097]	lost
natural/istic_MSK_dino518_ repro/natural_boot.json	all mask-erm	+0.016 [−0.007, +0.038]	lost
natural/istic_MSK_dino518_ repro/natural_boot.json	all balanced-mask	−0.018 [−0.041, +0.005]	lost
finetune/thyroid/resnet50	trapA mte_post mask_post	−0.000 [−0.001, +0.001]	lost
finetune/thyroid/resnet50	trapA cons_ft mask	+0.042 [+0.011, +0.074]	gained
finetune/thyroid/resnet50	trapB mte_ft mask	−0.042 [−0.089, +0.003]	lost
finetune/thyroid/resnet50	trapB mask_balanced balanced	+0.069 [−0.004, +0.143]	lost
finetune/thyroid/vit_small_ patch16_224.augreg_in21k_ft_ in1k	trapA mask_balanced balanced	+0.059 [+0.003, +0.122]	gained
finetune/thyroid/vit_small_ patch16_224.augreg_in21k_ft_ in1k	trapA umte_ft mte_ft	+0.032 [−0.072, +0.125]	lost
finetune/capsule/resnet50	trapB mte_ft erm	+0.120 [+0.020, +0.217]	gained
review2/transplant/ovary_dino518	0.0 balanced erm test_rev	+0.019 [−0.006, +0.046]	lost

6 What failed or changed

- **The transplant “harm” is mostly a paste effect.** Earlier drafts quoted the pasted-hair effect as the harm of masking with in-lesion hair; that attribution is withdrawn. Only N3 is attributed to the artifact, and it is significant for hair and thyroid only.
- **Capsule cannot be matched** under the registered rule; its crossover is descriptive.
- **The bootstrap correction** changed 179 verdicts, and the claims above were rewritten.

- **The capsule synthetic cohort did not reproduce its split**; its point estimates moved by up to 0.222, while the location interaction kept its sign and significance.

7 Conclusion

Supported. The real-artifact crossover is not explained by measured covariates: it survives matching (M1, 3 of 3 feasible cohorts) and doubly robust adjustment (A2, 3 of 3, Holm). Moving the same real artifact inside the ROI changes what masking does, and for hair and thyroid calipers the artifact’s content adds to the effect of the paste. **Qualified.** Most of the transplant interaction is produced by the paste itself; unmeasured confounding of the natural location cannot be excluded. **Not supported.** A causal harm figure for the real artifact from the transplant; the capsule matched analysis.

8 Questions for the supervisor

1. Is matching plus A2 plus the transplant sufficient causal evidence for a Q1 journal, or should an instrumental or negative-control analysis be attempted?
2. How should the paper present the paste effect: as a limitation of the transplant, or as a finding in itself (any label-correlated pattern inside the ROI defeats masking)?
3. Should the full old-versus-new interval table be published as supplementary material?

9 Next stage

Stage 5 moves from trap environments to practice: natural test sets without an imposed correlation, clinical operating points, the thyroid subgroup of 78 patients, the breadth of replication across encoders and fine-tuning, the DermLIP exception, and a theory judged by its held-out prediction error.

References

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- Meike Nauta, Ricky Walsh, Adam Dubowski, and Christin Seifert. Uncovering and correcting shortcut learning in machine learning models for skin cancer diagnosis. *Diagnostics*, 12(1):40, 2022.